More Than 67.32 Percent of the Zeta Zeros Are Simple and on the Critical Line

Deep Bhattacharjee [1,2,*] and Ushashi Bhattacharya [3,†]

[1] *Formerly, Electro-Gravitational Space Propulsion Laboratory (EGSPL), Bhubaneswar, Odisha 751030, India*
[2] *Formerly, University of Calcutta, Senate House, 87/1 College Street, Kolkata 700073, West Bengal, India*
[3] *Formerly, National Taiwan University, Taipei 106319, Taiwan*

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We prove that more than 67.32 percent of the nontrivial zeros of the Riemann zeta function, counted with multiplicity, are simple and lie on the critical line, and that the method cannot reach 67.36 percent. The new ingredient is a sharp bound for the largest eigenvalue of the overlap matrix of well-separated zeros, which lets long blocks of zeros be used in full.

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I. INTRODUCTION

Write $\rho = \beta + i\gamma$ for a nontrivial zero of the Riemann zeta function, let $N(T_1, T_2)$ be the number of zeros with $T_1 < \gamma \le T_2$, counted with multiplicity, and let $N_0^s(T_1, T_2)$ be the number of those zeros that are simple and lie on the critical line $\beta = \frac{1}{2}$. That a positive proportion of the zeros lie on the line is due to Selberg [19]. Levinson's method [15] gave one third, and the zeros it detects are simple [12]; Conrey [11] passed two fifths, Bui, Conrey and Young [9] passed 41 percent, and Pratt, Robles, Zaharescu and Zeindler [18] passed five twelfths.[1]

Montgomery [16] took a different route. Assuming the Riemann hypothesis, he evaluated the pair correlation of the zeros against test functions whose Fourier transform is supported in $[-1, 1]$, and deduced that at least two thirds of the zeros are simple. Montgomery and Taylor [17] optimised the test function and reached

$$H_{\mathrm{MT}} := \frac{3}{2} - \frac{1}{\sqrt{2}} \cot \frac{1}{\sqrt{2}} = 0.6725007036\ldots, \qquad (1)$$

which is optimal among certificates of this shape [10].[2]

In 2026 Alpöge and Furman [2] removed the Riemann hypothesis from Montgomery's deduction. They compress Weil's Hermitian form to a family of modulated copies of a window, split the resulting matrix into a positive part carried by the zeros on the line and a part of controlled positive index carried by the off-line pairs, and bound the number of simple zeros on the line by a rank–trace inequality; the analytic input is the unconditional pair correlation of Baluyot, Goldston, Suriajaya and Turnage-Butterbaugh [3]. This gives $\frac{2}{3}$, and H_{MT} with the Montgomery–Taylor window, unconditionally.[3] A refinement of the rank–trace inequality, which retains a nonnegative

* itsdeep@live.com; ∗Corresponding author. Second address: d.bhattacharjee@erl-forschung.de.
† rai.bhattacharya.in@gmail.com

[1] All of these results come from mollified moments of ζ or of a combination of its derivatives on the line. The mollifier is a Dirichlet polynomial chosen to cancel the large values of ζ, and the proportion grows with its length; the length is limited by the available estimates for sums of Kloosterman sums. Selberg's original argument already gave a positive proportion, without an explicit constant; Levinson's constant is slightly above 0.3474, and the improvements since then each came from a longer or more flexible mollifier.

[2] Montgomery's form factor $F(\alpha)$ equals $|\alpha| + o(1)$ for $|\alpha| \le 1$; the zero side of his identity is a sum over pairs of zeros weighted by a Fourier transform, and on the Riemann hypothesis every ordinate is real, so a positive test function gives a sum of positive terms. The window that attains H_{MT} is $\cos(\sqrt{2}t)$ on $[-\frac{1}{2}, \frac{1}{2}]$, and the constant is $2 - R(\psi)$, where $R(\psi)$ is the normalised second moment of the window; see [2, Lemma 5.6].

[3] The paper states that the argument was found and written by Claude, a language model developed by Anthropic, and that the authors verified it; it is accompanied by a Lean 4 formalisation. Lamzouri [14] gave a second proof of the bound $\frac{2}{3}$.

stability defect of the Gram matrix of the simple zeros, and a certified inequality for seven consecutive zeros, raised the constant to 0.6730085 [1]. Our result is the following.

Theorem 1. *As $T \to \infty$,*

$$\liminf_{T\to\infty} \frac{N_0^s(T,2T)}{N(T,2T)} \geq \frac{125000}{124111}\Big(H_{\mathrm{MT}} - \frac{7999}{2000000}\Big) = 0.6732896395\ldots.$$

Summing over dyadic intervals gives the same constant for $N_0^s(0,T)/N(0,T)$. Figure 1 places the bound among the earlier ones.

The gain over [1] comes from using the stability defect in full. For a positive semidefinite matrix M the defect is $\mathscr{D}(M) = \mathrm{tr}\,\Psi(M)$, with $\Psi(t) = (t-1)^2$ on $[0,2]$ and $\Psi(t) = 2t-3$ beyond. When every eigenvalue of M is at most 2, the defect equals $\mathrm{tr}(M-I)^2$, which for a Gram matrix with unit diagonal is twice the sum of the squared overlaps of the zeros. Without control of the eigenvalues, the defect of a block of zeros can only be bounded by 1, and in [1] this cap limits the size of the blocks and hence the gain. We prove that the overlap matrix of points at mutual distance at least $\frac{11}{20}$ of the mean spacing has every eigenvalue at most 1.97903 (Lemma 8). The bound is sharp: the eigenvalues of evenly spaced points reach it, and points spaced more closely than 0.5442 produce eigenvalues above 2. Zeros closer than $\frac{11}{20}$ are treated separately: they overlap so strongly that a pair of them alone has defect at least 0.719, far more than they cost.

With the cap removed, a block of m zeros has defect proportional to m, and the remaining question is the energy of ten consecutive zeros. Each pair of the ten zeros enters with a weight that depends on its place in the window, chosen by linear programming, and with these weights the inequality sees nearly all the energy that ten consecutive zeros carry. It is an inequality in nine variables, which we verify by branch and bound in outward-rounded arithmetic over tables of interval enclosures of the kernel and its first two derivatives (Proposition 11 and Appendix B). The extremal configurations are close to a lattice with gaps near 1.04 and 1.97, because the overlap kernel vanishes at 1.0573, 2.0301 and 3.0202.

The method has a limit. Periodic arrangements of zeros satisfy every constraint the argument uses, and their energy shows that no choice of window length or weights can push the bound past 0.67357 (Sec. IX). This is well below the ceiling 0.6818 that Alpöge and Furman identify for all certificates based on pair correlation with Fourier support in $[-1,1]$ [2, §7.2].

The route to this argument was found by the learn-from-failure protocol of Anaxia, the BLLM engine [6, 8].[4] For other approaches to the zeros through spectral and large-sieve methods, see [5, 7].

The paper is organised as follows. Section II recalls the framework of [2]. Section III proves the stability form of the rank–trace inequality and the pinching inequality. Section IV computes the overlap kernel, and Sec. V proves the eigenvalue bound. Section VI bounds the defect of a block of zeros, Sec. VII states the ten-point inequality, and Sec. VIII proves Theorem 1. Section IX shows where the method stops. Appendix A gives the elementary estimates in detail, and Appendix B describes the verification.

II. THE FRAMEWORK

We use the notation of Alpöge and Furman [2]. Let

$$L := \log\frac{T}{2\pi}, \qquad h := \frac{2\pi}{L},$$
$$I := [T,2T), \qquad I' := [T-\sqrt{T}, 2T+\sqrt{T}),$$

and let $\psi(t) = \cos(\sqrt{2}t)\,\mathbf{1}_{[-1/2,1/2]}(t)$ be the Montgomery–Taylor window. Fix a smooth nondecreasing χ with $\chi = 0$ on $(-\infty,0]$ and $\chi = 1$ on $[1,\infty)$, and set

$$\phi(u) := \chi\big(\tfrac{L}{2}+u\big)\chi\big(\tfrac{L}{2}-u\big)\,\psi(u/L)^{1/2}, \qquad a := \frac{\|\phi\|_2^2}{L}.$$

With $\widehat{f}(\xi) = \int f(u)e^{-iu\xi}\,du$, the grid points $\alpha_k := T + kh$, and $d := \lfloor LT/2\pi \rfloor$, each zero ρ gives a vector

$$v_\rho := \big(\widehat{\phi}(\gamma_\rho - \alpha_k)\big)_{0\leq k<d}, \qquad \tilde{v}_\rho := \frac{v_\rho}{\sqrt{a}L}.$$

The matrix $\widetilde{G} := \sum_\rho m_\rho \tilde{v}_\rho \tilde{v}_\rho^{\mathsf{T}}$, the sum over the zeros with real part of γ_ρ in I', is the compression of Weil's explicit formula to the modulated copies of ϕ.[5] Let s_1 be the number of simple zeros on the line, s_2 the number of distinct multiple zeros on the line, and p the number of pairs $\{\rho, 1-\bar{\rho}\}$ off the line, all with real part of the ordinate in I'. Let V be the $d \times s_1$ matrix whose columns are the vectors $\tilde{v}_\rho$ of the simple zeros on the line, and put

$$P_1 := VV^{\mathsf{T}}, \qquad Q' := \widetilde{G} - P_1, \qquad M := V^{\mathsf{T}}V.$$

[4] The protocol records each attempt that fails together with the rule it teaches, and requires every later attempt to respect the rules so far. Eight rounds preceded this paper. The direct attacks on the Riemann hypothesis failed and were recorded as such: a correlation bound below the square-root size of each shift, a Nyman–Beurling completeness argument that uses only integer dilations, and shift-by-shift Chowla bounds. Two rules then shaped the present work. The first was that information beyond pair correlation can be used only where off-line zeros are provably rare; three-point correlations raise the ceiling of the method to about 0.709, but with the zero-density estimates now known only a gain of 0.0002 survives, which closed that route. The second was that a capped block defect saturates near 0.67304, which pointed to the uncapped defect and to the eigenvalue bound of Lemma 8. The engine did not produce the proofs; the failure log, the rules and every computation are in the archive of this paper (see the data availability section).

[5] Weil showed that the Riemann hypothesis is equivalent to the positivity of the Hermitian form $W(f,g) = \sum_\rho m_\rho \widehat{f}(\gamma_\rho)\overline{\widehat{g}(\gamma_\rho)}$ on compactly supported test functions [22]. Off the line the ordinates γ_ρ are complex, and a pair of zeros $\rho, 1-\bar{\rho}$ contributes an indefinite form of signature $(1,1)$. The argument of [2] needs no positivity: it counts the positive directions instead.

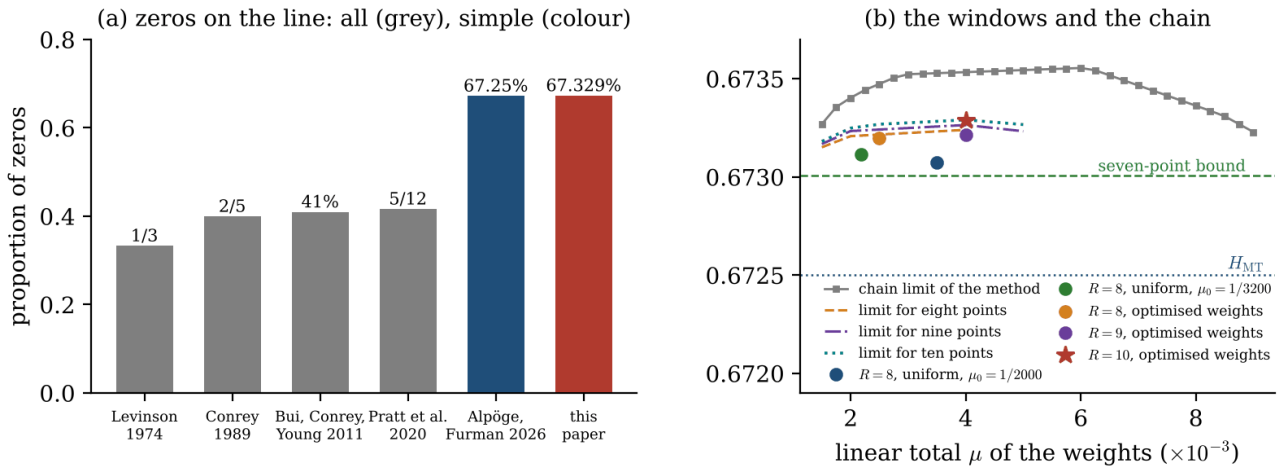


FIG. 1. (a) Unconditional lower bounds for the proportion of zeros on the critical line. The grey bars are the proportions of zeros on the line proved by mollifiers, in the form stated in the titles of the papers; the coloured bars are proportions of zeros that are both simple and on the line. (b) The bound $(H_{\mathrm{MT}}-\mu)/(1-c_0)$ from an inequality for R consecutive zeros, against the linear total μ of its weights: the certified points for $R=8$ with uniform and with optimised weights, for $R=9$ with optimised weights and for $R=10$ with the weights of Table I, the seven-point bound of [1], the Montgomery–Taylor constant, the limits for eight, nine and ten points, and the upper limit of the method obtained from periodic configurations (Sec. IX).

Proposition 2 ([2]). *As $T\to\infty$:*

(a) every column of V has norm at most 1, $n_+(Q')\le s_2+p$, and $s_1+2s_2+2p\le N(I')$;

(b) $\operatorname{tr}\widetilde{G}=N(T,2T)+o(N(T,2T))$;

(c) $\|\widetilde{G}\|_{\mathrm{HS}}^2=(2-H_{\mathrm{MT}}+o(1))N(T,2T)$;

(d) for all $z,z'\in\mathbb{C}$, $\sum_{k\in\mathbb{Z}}\widehat{\phi}(z-\alpha_k)\widehat{\phi}(z'-\alpha_k)=L\widehat{\phi^2}(z-z')$, and $|\widehat{\phi}(r)|\ll\min(L,r^{-2})$ for real r.

Proof. Part (a) is Lemma 2.1 at $z=z'=\gamma_\rho$, Proposition 4.1 and the decomposition in the proof of Theorem A in [2]. Part (b) is Proposition 4.2 there together with $N(I')-N(I)\ll\sqrt{T}\log T$. Part (c) is Theorem 5.7 with Lemma 5.6, since $R(\psi)=2-H_{\mathrm{MT}}$ for the Montgomery–Taylor window. Part (d) is Lemma 2.1 and the bound (2.8) of [2]. □

Here n_+ is the number of positive eigenvalues and $\|X\|_{\mathrm{HS}}^2=\operatorname{tr}X^2$. Proposition 2(b) and (c) are where the prime numbers enter: (c) is Montgomery's second moment, evaluated without the Riemann hypothesis [3].

III. THE STABILITY DEFECT

Define the convex function

$$\Psi(t):=\begin{cases}(t-1)^2, & 0\le t\le 2,\\ 2t-3, & t\ge 2,\end{cases}$$

and, for a positive semidefinite matrix X, the stability defect $\mathscr{D}(X):=\operatorname{tr}\Psi(X)\ge 0$. The function Ψ is continuously differentiable, with $\Psi'(2)=2$ from both sides.

Lemma 3 (Rank–trace inequality with defect). *Let V be a $d\times r$ matrix whose columns have norm at most 1, put $P=VV^*$ and $M=V^*V$, and let Q be Hermitian with $n_+(Q)\le b$. Then*

$$\|P+Q\|_{\mathrm{HS}}^2\ge 4\operatorname{tr}(P+Q)-3r-4b+\mathscr{D}(M).$$

The case $\mathscr{D}(M)=0$ is the rank–trace inequality of [2, Lemma 3.2]; the defect term is the refinement of [1]. We give the proof.

Proof. Write $Q=Q_+-Q_-$ with $Q_\pm\succeq 0$, $Q_+Q_-=0$ and $\operatorname{rank}Q_+\le b$. Since $\operatorname{tr}(PQ_+)\ge 0$ and $\operatorname{tr}(Q_+Q_-)=0$,

$$\|P+Q\|_{\mathrm{HS}}^2\ge\|P-Q_-\|_{\mathrm{HS}}^2+\|Q_+\|_{\mathrm{HS}}^2.$$

Every positive eigenvalue q of Q_+ satisfies $q^2\ge 4q-4$, so $\|Q_+\|_{\mathrm{HS}}^2\ge 4\operatorname{tr}Q_+-4b$. Let $p_1\ge\cdots\ge p_d\ge 0$ be the eigenvalues of P; the first r of them are the eigenvalues of M and the others vanish. Let $n_1\ge\cdots\ge n_d\ge 0$ be the eigenvalues of Q_-. Von Neumann's trace inequality $\operatorname{tr}(PQ_-)\le\sum_i p_in_i$ gives

$$\|P-Q_-\|_{\mathrm{HS}}^2+4\operatorname{tr}Q_-\ge\sum_{i\le r}\big((p_i-n_i)^2+4n_i\big),$$

because the terms with $i>r$ are $n_i^2+4n_i\ge 0$. For $p\ge 0$,

$$\min_{n\ge 0}\big((p-n)^2+4n\big)=2p-1+\Psi(p),$$

the minimum being at $n=0$ when $p\le 2$ and at $n=p-2$ when $p\ge 2$.[6] Hence $\|P-Q_-\|_{\mathrm{HS}}^2\ge 2\operatorname{tr}P-r+\mathscr{D}(M)-4\operatorname{tr}Q_-$.

[6] For $p\le 2$ the minimum is $p^2=2p-1+(p-1)^2$, and for $p\ge 2$ it is $4+4(p-2)=4p-4=2p-1+(2p-3)$. The function Ψ is thus the exact loss in the scalar inequality $p^2\ge 2p-1$ when one allows a compensating negative eigenvalue; this is what makes the defect the right quantity to keep.

Adding the two bounds,

$$\|P+Q\|_{\mathrm{HS}}^2 \ge 4\,\mathrm{tr}(P+Q) - 2\,\mathrm{tr}\,P - r - 4b + \mathscr{D}(M),$$

and $\mathrm{tr}\,P = \sum \|v_j\|^2 \le r$. □

Lemma 4 (Pinching)**.** *Let $X \succeq 0$ be an $n\times n$ matrix and let $B_1,\dots,B_q$ be a partition of $\{1,\dots,n\}$. Then $\mathscr{D}(X) \ge \sum_j \mathscr{D}(X_{B_jB_j})$, where X_{BB} is the principal submatrix on B.*

Proof. Let Π_j be the coordinate projection onto B_j and, for $\varepsilon \in \{\pm 1\}^q$, let $U_\varepsilon = \sum_j \varepsilon_j \Pi_j$. Then $\sum_j \Pi_j X \Pi_j = 2^{-q}\sum_\varepsilon U_\varepsilon X U_\varepsilon$. The map $Y \mapsto \mathrm{tr}\,\Psi(Y)$ is convex on positive semidefinite matrices, since Ψ is convex on $[0,\infty)$ [4, Theorem V.2.5], and it is unitarily invariant. Hence $\mathscr{D}(\sum_j \Pi_j X \Pi_j) \le \mathscr{D}(X)$, and the left side is $\sum_j \mathscr{D}(X_{B_jB_j})$. □

Corollary 5. *Let M° be the principal submatrix of M indexed by any set of simple zeros on the line with ordinates in I'. Then*

$$N_0^s(T,2T) \ge H_{\mathrm{MT}} N(T,2T) + \mathscr{D}(M^\circ) - o(N(T,2T)).$$

Proof. Apply Lemma 3 to $P_1 + Q' = \widetilde{G}$ with $r = s_1$ and $b = s_2 + p$ (Proposition 2(a)):

$$\begin{aligned}\|\widetilde{G}\|_{\mathrm{HS}}^2 &\ge 4\,\mathrm{tr}\,\widetilde{G} - 3s_1 - 4s_2 - 4p + \mathscr{D}(M)\\ &\ge 4\,\mathrm{tr}\,\widetilde{G} - s_1 - 2N(I') + \mathscr{D}(M).\end{aligned}$$

By Proposition 2(b), (c) and $N(I') = N(T,2T) + o(N(T,2T))$, this is $s_1 \ge H_{\mathrm{MT}} N(T,2T) + \mathscr{D}(M) - o(N(T,2T))$. Lemma 4, applied to M° and its complement, gives $\mathscr{D}(M) \ge \mathscr{D}(M^\circ)$, and $N_0^s(T,2T) \ge s_1 - O(\sqrt{T}\log T)$. □

IV. THE OVERLAP KERNEL

Put

$$K(x) := \int_{-1/2}^{1/2} \cos(\sqrt{2}t)\cos(2\pi x t)\,dt,$$
$$k(x) := \frac{K(x)}{K(0)}, \qquad w := k^2,$$

so that $K(0) = K_0 := \sqrt{2}\sin(1/\sqrt{2}) = 0.9187253698\dots$ and

$$K(x) = \frac{1}{2}\Big(\mathrm{sinc}\big(\pi x - \tfrac{1}{\sqrt{2}}\big) + \mathrm{sinc}\big(\pi x + \tfrac{1}{\sqrt{2}}\big)\Big), \tag{2}$$

with $\mathrm{sinc}\,s = (\sin s)/s$. For a simple zero ρ on the line put $x_\rho := (\gamma_\rho - T)/h$, its ordinate measured in units of h, which is the mean spacing of the zeros up to a factor $1 + O(1/L)$.

Lemma 6. *Retain the simple zeros on the line with $x_\rho \in [L^2, d - L^2]$. The others number $o(N(T,2T))$ among those in I, and for every fixed $S > 0$, uniformly for retained zeros with $|x_\rho - x_{\rho'}| \le S$,*

$$\langle \tilde{v}_\rho, \tilde{v}_{\rho'}\rangle = k(x_\rho - x_{\rho'}) + o(1).$$

Proof. By Proposition 2(d), the sum over all $k \in \mathbb{Z}$ equals $(aL)^{-1}\widehat{\phi^2}(hx)$ with $x = x_\rho - x_{\rho'}$. The grid points outside $[0,d)$ lie at distance at least $hL^2 = 2\pi L$ from γ_ρ and $\gamma_{\rho'}$, so by Cauchy–Schwarz and the decay of $\widehat{\phi}$ they contribute $\ll L^{-2}\sum_{j\ge 0}(2\pi L + jh)^{-4} \ll L^{-4}$ after division by aL^2. For the full sum, the substitution $u = Lt$ gives

$$\frac{\widehat{\phi^2}(hx)}{aL} = \frac{\int \phi(Lt)^2 e^{-2\pi i x t}\,dt}{\int \phi(Lt)^2\,dt}.$$

As $T \to \infty$, $\phi(Lt)^2 \to \psi(t)$ in L^1, because the two factors χ differ from 1 only on a set of t of measure $O(1/L)$. The ratio therefore tends to $K(x)/K_0$, uniformly for $|x| \le S$. The deleted zeros have ordinates in two intervals of length $2\pi L$ at the ends of I, and there are $O(L^2)$ of them by $N(t,t+1) \ll \log t$ [20, Theorem 9.2]. □

The kernel has three properties that the argument uses (Fig. 2).

Lemma 7. *(i) $k(0) = 1$, $|k| \le 1$, and k is decreasing on $[0,1]$.*

(ii) $w(x) \ge w(\frac{11}{20}) = 0.3595640596\dots$ for $0 \le x \le \frac{11}{20}$.

(iii) The first three positive zeros of k are $1.0572782\dots$, $2.0300675\dots$ and $3.0202429\dots$, and $w(x) = O(x^{-2})$.

Proof. (i) $|K(x)| \le \int \cos(\sqrt{2}t)\,dt = K_0$. For $0 < x \le 1$ and $0 < t \le \frac{1}{2}$ one has $0 < 2\pi x t \le \pi$, so

$$K'(x) = -\int_{-1/2}^{1/2} 2\pi t \cos(\sqrt{2}t)\sin(2\pi x t)\,dt < 0.$$

(ii) follows from (i) and $k(\frac{11}{20}) > 0$; the value is computed in Appendix A. (iii) The zeros are those of (2); the decay follows from the same formula. □

V. SEPARATED ZEROS

Lemma 8. *Let $\delta_1 = \frac{11}{20}$ and let $x_1,\dots,x_n$ be real numbers with $|x_i - x_j| \ge \delta_1$ for $i \ne j$. Then every eigenvalue of the matrix $\big(k(x_i - x_j)\big)_{i,j}$ lies in*

$$\Big[0,\ \frac{1}{\delta_1 K_0}\Big] = [0,\ 1.9790264\dots].$$

Proof. For a real vector c, with $S(t) = \sum_j c_j e^{2\pi i x_j t}$,

$$\sum_{i,j} c_i c_j k(x_i - x_j) = \frac{1}{K_0}\int_{-1/2}^{1/2} \cos(\sqrt{2}t)\,|S(t)|^2\,dt \ge 0,$$

because the cosine is positive on $[-\frac{1}{2},\frac{1}{2}]$. Let $\Lambda(t) = \mathrm{sinc}^2(\pi\delta_1 t)$. Its Fourier transform is the triangle

$$\int_{\mathbb{R}} \Lambda(t) e^{2\pi i \xi t}\,dt = \frac{1}{\delta_1}\Big(1 - \frac{|\xi|}{\delta_1}\Big)_+,$$

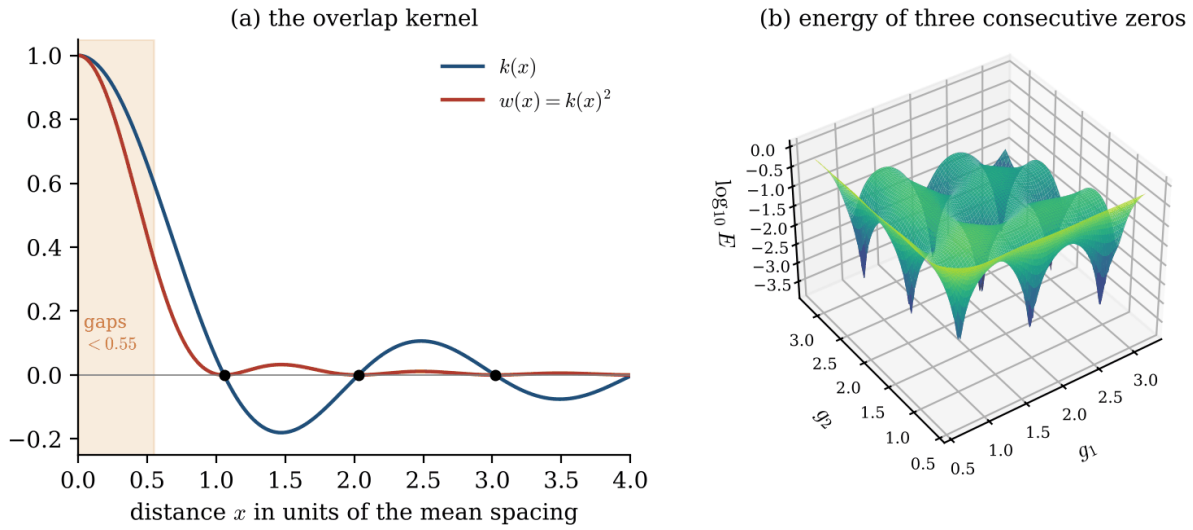


FIG. 2. (a) The overlap kernel k and its square w, in units of the mean spacing; the dots are the zeros of k, near 1, 2 and 3. The shaded strip marks the gaps below $\frac{11}{20}$, where the eigenvalue bound of Lemma 8 no longer applies and zeros are treated as clusters. (b) The energy $E = 2[w(g_1)+w(g_2)+w(g_1+g_2)]$ of three consecutive zeros with gaps g_1, g_2, on a logarithmic scale. Its valleys follow the zeros of k; the extremal configurations of Proposition 11 sit in these valleys, with gaps near 1.04 and 1.97.

which vanishes for $|\xi| \geq \delta_1$. We show in Appendix A that

$$\Lambda(t) \geq \cos(\sqrt{2}t) \qquad \text{for } |t| \leq \tfrac{1}{2}. \tag{3}$$

Since $\Lambda \geq 0$ everywhere, Λ majorises $\cos(\sqrt{2}t)\mathbf{1}_{[-1/2,1/2]}$ on the whole line, so

$$\int_{-1/2}^{1/2} \cos(\sqrt{2}t)|S|^2\,dt \leq \int_{\mathbb{R}} \Lambda|S|^2\,dt = \sum_{i,j} c_i c_j \widehat{\Lambda}(x_i - x_j) = \frac{\|c\|^2}{\delta_1},$$

because $\widehat{\Lambda}(x_i - x_j) = 0$ for $i \neq j$.[7] Dividing by K_0 gives the upper bound. □

The constant is sharp. For the points $x_j = j\delta$, $1 \leq j \leq n$, and the vector of ones, the Rayleigh quotient is $\sum_{|\ell|<n}(1-|\ell|/n)\,k(\ell\delta)$, the n-th Fejér mean at 0 of the Fourier series of the 1-periodic function $\theta \mapsto (\delta K_0)^{-1}\sum_{j\in\mathbb{Z}} \psi((j-\theta)/\delta)$ by Poisson summation. For $\delta < 2$ only the term $j = 0$ is nonzero near $\theta = 0$, the function is continuous there with value $1/(\delta K_0)$, and Fejér's theorem gives $\lambda_{\max} \geq 1/(\delta K_0) - o(1)$ as $n \to \infty$. So lattices with spacing below the critical value

$$\delta^* = \frac{1}{2K_0} = 0.5442322\ldots$$

have eigenvalues above 2, and Lemma 8 cannot hold with $\frac{1}{2}$ in place of $\frac{11}{20}$. Figure 3 shows the majorant and the eigenvalues of lattices.

VI. THE DEFECT OF A BLOCK OF ZEROS

Fix an integer $R \geq 2$. A system of weights for R points consists of numbers $b_1, \ldots, b_{R-1} > 0$ and $a_{r,i} \geq 0$, for $1 \leq r \leq R-1$ and $0 \leq i \leq R-1-r$, such that

$$\sum_{i=1}^{R-1} b_i = \mu, \qquad \sum_{i=0}^{R-1-r} a_{r,i} \leq 2 \quad (1 \leq r \leq R-1); \tag{4}$$

we call μ the linear total of the weights. For gaps $g = (g_1, \ldots, g_{R-1})$ in $[0,\infty)^{R-1}$ put $y_0 = 0$, $y_i = g_1 + \cdots + g_i$, and

$$F(g) := \sum_{i=1}^{R-1} b_i g_i + \sum_{r=1}^{R-1}\sum_{i=0}^{R-1-r} a_{r,i}\, w(y_{i+r} - y_i). \tag{5}$$

The second sum runs over all pairs of the R points $y_0 < \cdots < y_{R-1}$, and the pair (y_i, y_{i+r}) carries the weight $a_{r,i}$, which depends on its place in the window. The uniform weights $b_i = \mu_0$, $a_{r,i} = 2/(R-r)$ give the function F_R used in [1], with $\mu = (R-1)\mu_0$.[8]

[7] This is the mechanism of the large sieve: a band-limited majorant of the weight turns a quadratic form over separated points into its diagonal. Beurling and Selberg chose the majorant to optimise the large sieve constant; see [21]. Here the weight is $\cos(\sqrt{2}t)$ on an interval of length 1 rather than the indicator, and the separation δ_1 is below 1, the range in which the classical constant $1 + 1/\delta$ exceeds 2 and is useless for us. The Fejér kernel Λ with exactly the bandwidth δ_1 wins because $\cos(\sqrt{2}t)$ is flat to second order at $t = 0$, where Λ has curvature $-(\pi\delta_1)^2/3 \approx -0.995$, almost the -1 of the cosine.

[8] Every system of weights gives the same totals in the proof of Lemma 9; what the weights change is the minimum of a single window. The extremal

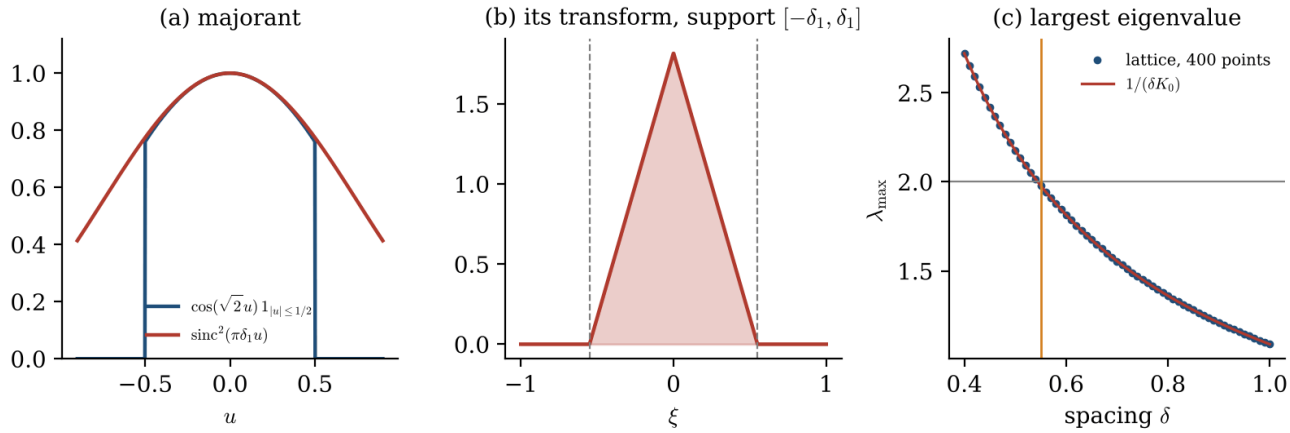


FIG. 3. The eigenvalue bound. (a) The Fejér kernel $\mathrm{sinc}^2(\pi\delta_1 u)$ lies above the Montgomery–Taylor window on $[-\frac{1}{2},\frac{1}{2}]$; the two curves agree to second order at $u=0$. (b) Its Fourier transform, a triangle supported in $[-\delta_1,\delta_1]$, so that points at distance at least δ_1 do not interact. (c) The largest eigenvalue of the overlap matrix of 400 evenly spaced points against the spacing δ, with the bound $1/(\delta K_0)$; the vertical line is $\delta_1 = \frac{11}{20}$ and the horizontal line is the threshold 2 of the defect.

Lemma 9. *Suppose that $F \ge c_0$ on $[0,\infty)^{R-1}$ for a system of weights with linear total μ. Then for all real $y_1 < \cdots < y_n$,*

$$2\sum_{i<j} w(y_j - y_i) + \mu\,(y_n - y_1) \ge c_0\,(n - R + 1).$$

Proof. For $n < R$ the right side is not positive. For $n \ge R$, add the inequality $F \ge c_0$ over the $n-R+1$ windows $y_s, \dots, y_{s+R-1}$ of R consecutive points. A pair y_j, y_{j+r} with $r < R$ sits at the place $i = j - s$ of the window that starts at s, and different windows give different places, so its total weight is at most $\sum_i a_{r,i} \le 2$; pairs at least R steps apart do not occur and $w \ge 0$. In the same way the gap $y_{j+1} - y_j$ is the i-th gap of at most one window for each i, so the linear terms add up to at most $\mu(y_n - y_1)$. □

For a set B of retained simple zeros, let G_B be its Gram matrix $(\langle \tilde v_\rho, \tilde v_{\rho'}\rangle)_{\rho,\rho'\in B}$ and let $\mathrm{span}(B)$ be the difference between the largest and the smallest x_ρ in B.

Proposition 10. *Suppose that $F \ge c_0$ on $[0,\infty)^{R-1}$ for a system of weights with linear total μ, and that*

$$(R+1)\,c_0 \le \tfrac{4}{3}\,w(\tfrac{11}{20}). \tag{6}$$

Fix $m \ge R$. Then, uniformly over all sets B of m consecutive retained simple zeros on the line,

$$\mathscr{D}(G_B) + \mu\,\mathrm{span}(B) \ge c_0\,(m - R + 1) - o(1).$$

Proof. If $\mu\,\mathrm{span}(B) \ge c_0(m-R+1)$ there is nothing to prove, since $\mathscr{D} \ge 0$. Otherwise $\mathrm{span}(B) < S := c_0(m-R+1)/\mu$, so by Lemma 6 every entry of G_B is $k(x_\rho - x_{\rho'}) + o(1)$, uniformly in B. Write $y_1 < \cdots < y_m$ for the points x_ρ, $\rho \in B$.

Call a gap $y_{i+1} - y_i$ short if it is less than $\delta_1 = \frac{11}{20}$. A cluster is a maximal set of at least two consecutive points joined by short gaps; let there be q clusters, of sizes $s_1, \dots, s_q \ge 2$. The points outside the clusters fall into at most $q+1$ maximal sets of consecutive points, the runs, of sizes $n_1, n_2, \dots$; inside a run every gap is at least δ_1, so all distances are. Split each cluster into $\lfloor s_j/2 \rfloor$ pairs of consecutive points and, when s_j is odd, one single point. Runs, pairs and single points partition B (Fig. 4), and Lemma 4 gives

$$\mathscr{D}(G_B) \ge \sum_{\text{runs}} \mathscr{D}(G_{\text{run}}) + \sum_{\text{pairs}} \mathscr{D}(G_{\text{pair}}).$$

The function $X \mapsto \mathscr{D}(X)$ is uniformly continuous on the matrices of size at most m with entries bounded by 2, so in each term we may replace G by the limiting matrix $(k(y_i - y_j))$ at a cost $o(1)$. For a run, the limiting matrix has unit diagonal and, by Lemma 8, all its eigenvalues in $[0,2]$; hence its defect is $\mathrm{tr}(X-I)^2$, which is twice the sum of w over the pairs of points of the run. For a pair at distance $g < \delta_1$ the limiting matrix has eigenvalues $1 \pm k(g) \in [0,2]$, and its defect is $2w(g) \ge 2w(\delta_1)$ by Lemma 7(ii). With Lemma 9 for each run and $\mathrm{span}(B) \ge \sum \mathrm{span}(\text{run})$,

$$\begin{aligned}
\mathscr{D}(G_B) + \mu\,\mathrm{span}(B) &\ge c_0 \sum_{\text{runs}} (n - R + 1) - o(1) \\
&\quad + 2w(\delta_1) \sum_j \Big\lfloor \frac{s_j}{2} \Big\rfloor \\
&\ge c_0(m - R + 1) - o(1) \\
&\quad + \sum_{j=1}^{q} \Big(2w(\delta_1) \Big\lfloor \frac{s_j}{2} \Big\rfloor - c_0 s_j \\
&\qquad - c_0(R-1) \Big),
\end{aligned}$$

using $\sum n + \sum s_j = m$ and at most $q+1$ runs. Since $\lfloor s/2 \rfloor \ge s/3$

configurations of F_R do not use all places alike: some pairs sit near a zero of k at every near-minimal configuration and contribute almost nothing, while others always carry a visible overlap. Moving weight from the first kind of place to the second raises the minimum.

TABLE I. The weights of Proposition 11: the pair weights $a_{r,i}$ in units of 10^{-5} (row r, place i) and the linear weights b_{i+1} in units of 10^{-7}. Both are unchanged when the window is reversed, $a_{r,i} = a_{r,R-1-r-i}$ and $b_i = b_{R-i}$.

	$i=0$	1	2	3	4	5	6	7	8
$a_{1,i}$	11763	20184	23216	29258	31155	29258	23216	20184	11763
$a_{2,i}$	0	23331	24628	52039	52039	24628	23331	0	
$a_{3,i}$	0	0	70557	58885	70557	0	0		
$a_{4,i}$	100000	0	0	0	0	100000			
$a_{5,i}$	32246	67753	0	67753	32246				
$a_{6,i}$	100000	0	0	100000					
$a_{7,i}$	0	200000	0						
$a_{8,i}$	100000	100000							
$a_{9,i}$	200000								
b_{i+1}	2611	4430	4546	5618	5585	5618	4546	4430	2611

for $s \geq 2$, and $2w(\delta_1)/3 > c_0$ by (6), each bracket is at least $s_j(\frac{2}{3}w(\delta_1) - c_0) - c_0(R-1) \geq \frac{4}{3}w(\delta_1) - (R+1)c_0 \geq 0$. □

VII. TEN CONSECUTIVE ZEROS

Proposition 11. *Let $R = 10$ and let F be the function (5) with the weights of Table I, whose linear total is $\mu = \frac{7999}{2000000}$. Then*

$$F(g) \geq \frac{889}{125000} = 0.007112 \qquad \text{for all } g \in [0,\infty)^9.$$

The inequality is verified by a computation in rigorous arithmetic, described in full in Appendix B. The search covers the whole of $[0,\infty)^9$: the linear term alone reaches 0.007112 once $\sum b_i g_i \geq \frac{889}{125000}$, and on the remaining simplex the domain is subdivided into boxes of grid cells of width $\frac{1}{4000}$ until a lower bound computed from interval enclosures of w and of its first two derivatives settles each box.

The weights were chosen by linear programming. For a finite set of gap vectors, maximising the smallest value of F on the set over all systems of weights with linear total μ is a linear programme. Starting from the uniform weights, we added the local minimisers of the current F to the set and solved the programme again, for 46 rounds, until the bound it gives came within 10^{-6} of the limit for ten points (Sec. IX); the weights were then rounded down to the decimals of Table I, which keeps (4). The programme spreads the minimum over many configurations: a local minimisation from many starting points finds the smallest value 0.00711257, attained to within 10^{-7} at two different gap vectors and to within 10^{-6} at six (Fig. 5), among them

$$(1.038, 1.974, 1.040, 1.974, 1.041, 1.974, 1.040, 1.974, 1.038),$$

so the target $\frac{889}{125000}$ lies within 6×10^{-7} of the minimum; the next decimal, 0.0071125, is out of reach of cells of width $\frac{1}{4000}$. With uniform weights and the same μ the smallest value would be 0.0069096, which gives only 0.6731519.[9]

For nine points the same procedure gives weights with $\mu = \frac{19999}{5000000}$ for which the verifier certifies $F \geq \frac{7}{1000}$, and hence the bound $\frac{1000}{993}(H_{\mathrm{MT}} - \frac{19999}{5000000}) = 0.6732133\ldots$; for eight points it gives weights with $\mu = \frac{12499}{5000000}$ and $F \geq \frac{19}{4000}$, and the bound $\frac{4000}{3981}(H_{\mathrm{MT}} - \frac{12499}{5000000}) = 0.6731985\ldots$. With uniform weights the same verifier certifies $F_8 \geq \frac{13}{3125}$ for $\mu_0 = \frac{1}{3200}$ and $F_8 \geq \frac{605}{100000}$ for $\mu_0 = \frac{1}{2000}$, which give the weaker bounds $0.6731133\ldots$ and $0.6730727\ldots$ by the argument of Sec. VIII.

The inequality of [1] is the case $R = 7$ with uniform weights, $\mu_0 = \frac{1}{3000}$, $c_0 = \frac{19}{5000}$. There the defect of a whole block was capped at 1, so blocks could contain at most about $1/c_0$ windows, and a longer window gains nothing. Lemma 8 removes the cap, so the size of the blocks can tend to infinity and the window length can be chosen freely.

VIII. PROOF OF THEOREM 1

Take $R = 10$ and the weights of Proposition 11, so that $\mu = \frac{7999}{2000000}$, and $c_0 = \frac{889}{125000}$. Condition (6) holds, since $11c_0 = 0.078232 < 0.4794 < \frac{4}{3}w(\frac{11}{20})$. Fix $m \geq 10$ and let $x_1 < \cdots < x_{S^\circ}$ be the normalised ordinates of the retained simple zeros on the line, so that $S^\circ = N_0^s(T,2T) - o(N(T,2T))$ by Lemma 6. Let M° be their Gram matrix.

For each offset $j \in \{0,\ldots,m-1\}$, partition the points into the first j of them, consecutive blocks of m points, and a final piece of fewer than m points. By Lemma 4 and Proposition 10,

$$\mathscr{D}(M^\circ) \geq \sum_B \mathscr{D}(G_B) \geq \sum_B \big(c_0(m-R+1) - \mu\,\mathrm{span}(B) - o(1)\big).$$

For every offset there are at least $S^\circ/m - 2$ blocks and at most S°/m, and the $o(1)$ is uniform, so these terms total $o(N)$, with $N = N(T,2T)$. Average over the m offsets. A gap $x_{i+1} - x_i$ lies inside a block for at most $m-1$ of the m offsets, and $x_{S^\circ} - x_1 \leq d = N(1 + O(1/L))$, so

$$\mathscr{D}(M^\circ) \geq \frac{c_0(m-R+1)}{m}\,S^\circ - \frac{(m-1)\mu}{m}\,N - o(N).$$

Insert this in Corollary 5:

$$\Big(1 - \frac{c_0(m-R+1)}{m}\Big)N_0^s(T,2T) \geq \Big(H_{\mathrm{MT}} - \frac{(m-1)\mu}{m}\Big)N - o(N).$$

Divide by N and let $T \to \infty$, and then let $m \to \infty$:

$$\liminf_{T\to\infty} \frac{N_0^s(T,2T)}{N(T,2T)} \geq \frac{H_{\mathrm{MT}} - \mu}{1 - c_0} = \frac{125000}{124111}\Big(H_{\mathrm{MT}} - \frac{7999}{2000000}\Big).$$

The value is $0.6732896395962199\ldots$, computed in ball arithmetic [13]. □

[9] The extremal configuration is a piece of a near-lattice: gaps of about one and two mean spacings, each a little below the zero of k near 1 or the zero near 2, so that most overlaps are small and the remaining ones balance the linear term. Periodic arrangements of the same two gaps are the extremal chains of Sec. IX for values of μ near the one used here.

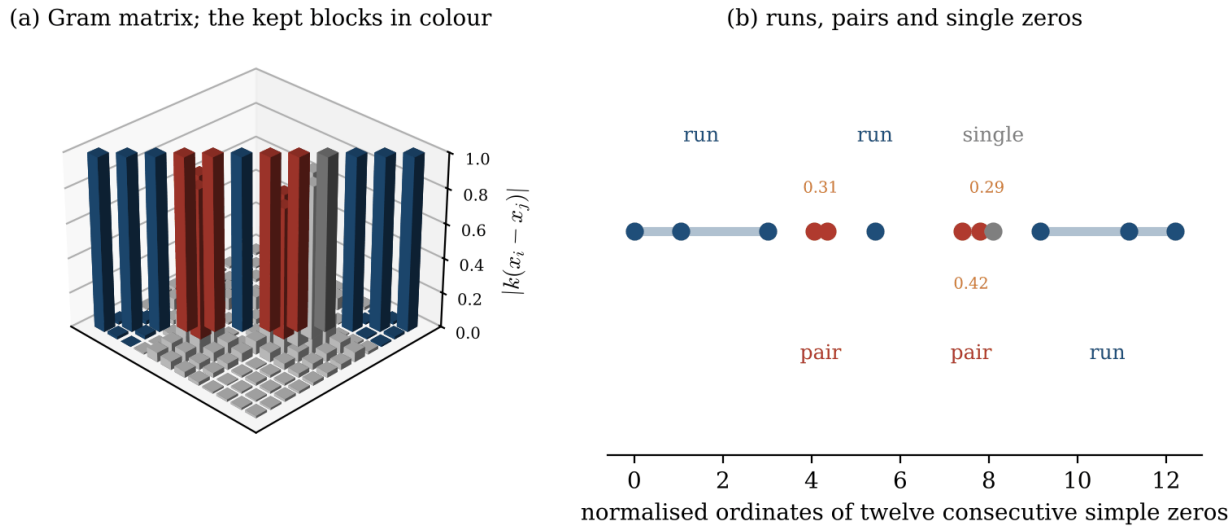


FIG. 4. The decomposition in the proof of Proposition 10, for twelve consecutive zeros with three short gaps (orange). (a) The Gram matrix; pinching keeps the coloured diagonal blocks and discards the grey entries. (b) The runs have all gaps at least $\frac{11}{20}$, so their eigenvalues stay below 2 and their defect is their full energy; the pairs joined by a short gap have defect at least 0.719 each, which pays for the windows that the clusters cut.

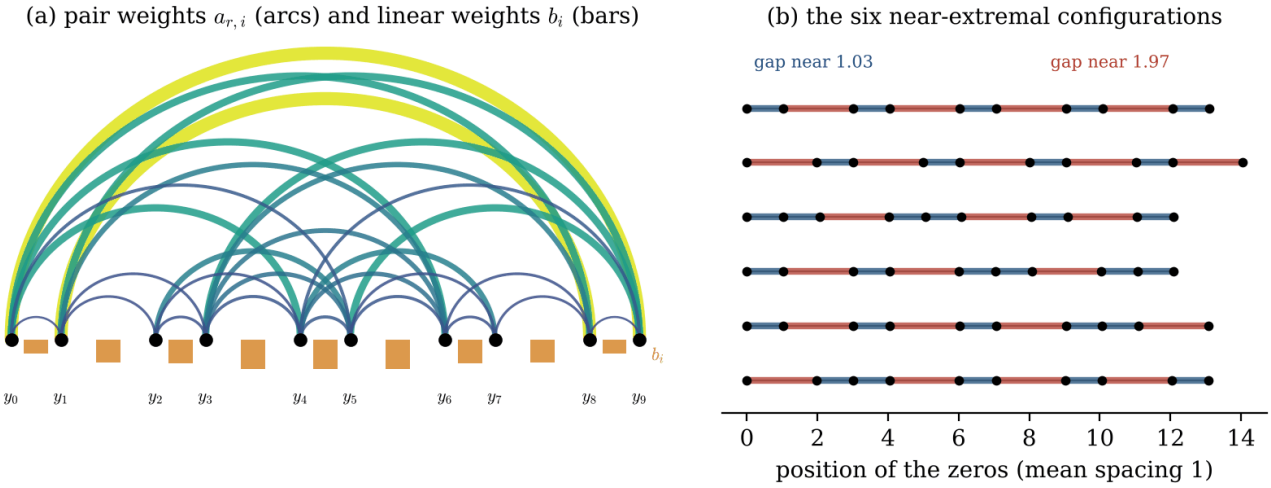


FIG. 5. (a) The weights of Table I over one of the extremal configurations of Proposition 11: each arc joins two of the ten points, with width proportional to its weight $a_{r,i}$, and the bars below show the linear weights b_i. The pair at the two ends and one pair seven steps apart carry the full weight 2, and fifteen pairs carry none. (b) The six gap vectors at which F comes within 10^{-6} of its smallest value 0.00711257. Every gap is close to 1.03 or to 1.97, a little below a zero of k, and the weights are balanced so that no single arrangement of the two gaps is cheaper than the others.

Remark 12. Every input of the proof is available for a Dirichlet L-function $L(s,\chi)$ with χ primitive: Proposition 2 holds there by [2, Theorem B], and Lemmas 3–9 do not involve ζ. The same constant therefore holds for the simple zeros of $L(s,\chi)$ on the critical line.

IX. HOW FAR THE METHOD GOES

The argument has three parameters: the window length R, the weights, with linear total μ, and the constant c_0, which is the minimum of F. Longer windows and better weights see more of the energy. With uniform weights the local minima of F_R give about 0.67309, 0.67313 and 0.67317 for $R = 7$, 8 and 9, while optimised weights reach 0.67321 with $R = 8$, 0.67326 with $R = 9$ and 0.67329 with $R = 10$. The following

TABLE II. The chain energy $e(\mu)$, the smallest value of $e(g,\mu)$ found over periodic gap patterns g of period at most 4, the pattern that attains it, and the resulting upper limit $(H_{\mathrm{MT}}-\mu)/(1-e(\mu))$ for the bound of Sec. VIII.

μ	$e(\mu)$	pattern	upper limit
0.0020	0.0043102	$(1.99, 1.98, 1.04, 1.98)$	0.6734032
0.0025	0.0051570	$(1.98, 1.04, 1.98)$	0.6734738
0.0030	0.0059725	$(1.04, 1.96)$	0.6735233
0.0040	0.0074727	$(1.04, 1.96)$	0.6735338
0.0050	0.0089729	$(1.04, 1.96)$	0.6735443
0.0060	0.0104731	$(1.04, 1.96)$	0.6735549
0.0065	0.0111605	$(1.02, 1.02, 1.95)$	0.6735175
0.0080	0.0131609	$(1.02, 1.02, 1.95)$	0.6733628

proposition bounds what any window and any weights can give.

Proposition 13. *Let $(g_i)_{i\in\mathbb{Z}}$ be a periodic sequence of positive gaps with period p and mean $\bar g$, let y be the corresponding points, and put*

$$e(g,\mu)=\mu\bar g+\frac{2}{p}\sum_{i=1}^{p}\sum_{j>i}w(y_j-y_i).$$

If $F\ge c_0$ holds for a system of weights with linear total μ, then $c_0\le e(g,\mu)$, and the bound obtained in Sec. VIII is at most $(H_{\mathrm{MT}}-\mu)/(1-e(g,\mu))$.

Proof. Apply Lemma 9 to n consecutive points of the sequence, divide by n and let $n\to\infty$; the double sum converges because $w(x)=O(x^{-2})$. □

Minimising $e(g,\mu)$ over periodic patterns of period at most 4 on a grid of weights gives the values in Table II. For a fixed pattern the upper limit $(H_{\mathrm{MT}}-\mu)/(1-e(g,\mu))$ is monotone in μ, since its derivative has the sign of $\bar g H_{\mathrm{MT}}+e(g,0)-1$, so the patterns found at the grid points bound it on every interval between them.[10] The resulting bound stays below 0.67357 for every $\mu\ge 0$, with the largest value near $\mu=0.0061$, where the best pattern is the alternation $(1.036, 1.964)$; for smaller μ gaps near 2 take over, and for larger μ patterns with more gaps near 1. So no window length or choice of weights improves Theorem 1 by more than 0.0004, and the approach cannot reach 67.36 percent. The proof of Proposition 13 uses only the pairs at most $R-1$ steps apart, so the double sum may be restricted to them; for $\mu=0.004$ the restricted chain energy gives the limits 0.6732646 for nine points and 0.6732908 for ten, and the weights of Table I come within 10^{-6} of the second before rounding. Longer windows gain slowly: at the same μ the limit is 0.6733072 for eleven points and 0.6733252 for twelve. Tilting the window, that is replacing $\cos(\sqrt{2}t)$ by $\cos(\alpha t)$, lowers H_{MT} faster than it raises the defect: at $R=7$ the best tilt, $\alpha=1.5$, gains 0.00004.

Going further needs information that the energy of runs does not see. Alpöge and Furman show that no certificate built from pair correlation with Fourier support in $[-1,1]$ can exceed about 0.6818 [2, §7.2]. Between 0.6736 and 0.6818 lies the part of that information which the stability defect, a function of the Gram matrix of the simple zeros alone, does not capture. Information beyond pair correlation can raise the ceiling: in the periodic models of [2, §7.2] with the three-point correlations of the circular unitary ensemble added, the ceiling rises to about 0.709. These correlations are known in the range used, but an off-line pair of zeros enters a triple product with a weight that grows faster than in the quadratic case, and the known zero-density estimates control only the triples whose frequencies add up to less than half of the range. With those alone the gain is about 0.0002.

None of this bears on the Riemann hypothesis itself. Every bound of this kind is insensitive to a set of zeros of density zero, and the argument applies equally to functions, such as the Davenport–Heilbronn function, whose zeros are known to leave the critical line.

DATA AND CODE AVAILABILITY

The proofs of Lemmas 3–9 and of Proposition 10 are by hand. Proposition 11 is verified by the rigorous computation described in Appendix B. The repository `creelie/mh-phys` contains the LaTeX source of this paper; the tables of interval enclosures of w and of its first two derivatives in 128-bit ball arithmetic, with their hashes; the branch-and-bound verifier, in serial, parallel, weighted and second-order versions, with the weights of Table I and of the shorter windows, the logs of the certified runs, and a workflow that rebuilds the tables and repeats the nine- and ten-point certificates on every change; independent checks in ball and floating-point arithmetic of the eigenvalue bound, of the endpoint values in Appendix A and of the eigenvalues of lattices; the local minimisations for $R=7,\dots,10$ and the linear programmes that choose the weights; the chain energies of Table II and their extension to every weight; the linear programmes behind the ceilings quoted in Sec. IX; and the failure log of the search that led to the argument. The archive is [23], doi:10.5281/zenodo.23489033, where later versions are listed. The search followed the learn-from-failure protocol of the BLLM engine Anaxia, release 3.1.1 [8]. Deep Bhattacharjee carried out the computations and the C, Python and shell checks with AI assistance from Claude (Anthropic); both authors approved the final text.

ETHICAL DECLARATIONS

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[10] The double sum is taken over all pairs: 2×10^5 neighbours are summed and the rest is bounded by $|k(x)|\le(K_0(\pi x-1/\sqrt{2}))^{-1}$. A truncation at a few hundred neighbours, which looks harmless because w is small there, lowers $e(g,\mu)$ by about 5×10^{-5}, more than the margins in the table.

Appendix A: Elementary estimates

1. The majorant

We prove (3). Both sides are even, so let $0 \le t \le \frac{1}{2}$ and $s = \pi\delta_1 t \in [0, 0.8640]$. The Taylor series

$$\operatorname{sinc}^2 s = \frac{1-\cos 2s}{2s^2} = \sum_{k\ge 1} (-1)^{k+1} \frac{2^{2k-1} s^{2k-2}}{(2k)!}$$

alternates, and for $s \le 1$ the ratio of consecutive terms is $4s^2/((2k+1)(2k+2)) < 1$, so the terms decrease in absolute value and the partial sum ending with a negative term is a lower bound:

$$\operatorname{sinc}^2 s \ge 1 - \frac{s^2}{3} + \frac{2s^4}{45} - \frac{s^6}{315}.$$

In the same way $\cos y \le 1 - \frac{y^2}{2} + \frac{y^4}{24}$ for all real y. With $y = \sqrt{2}t$ and $u = t^2 \in [0, \frac{1}{4}]$,

$$\Lambda(t) - \cos(\sqrt{2}t) \ge -u\,q(u),$$
$$q(u) = (\alpha_1 - 1) + \left(\tfrac{1}{6} - \alpha_2\right)u + \alpha_3 u^2,$$

where $\alpha_1 = (\pi\delta_1)^2/3 = 0.9951851\ldots$, $\alpha_2 = 2(\pi\delta_1)^4/45 = 0.3961573\ldots$ and $\alpha_3 = (\pi\delta_1)^6/315 = 0.0844821\ldots$. The quadratic q is convex, so its maximum on $[0, \frac{1}{4}]$ is at an endpoint, and

$$q(0) = -0.0048148\ldots < 0, \qquad q(\tfrac{1}{4}) = -0.0569074\ldots < 0.$$

Hence $\Lambda(t) \ge \cos(\sqrt{2}t)$ on $[0, \frac{1}{2}]$. The values are computed with π enclosed in 200-bit ball arithmetic; $q(0) = \alpha_1 - 1$ alone needs only $\pi^2 \cdot 0.3025 < 3$, that is $\pi^2 < 9.918$. The margin at small t is of order $0.0048\,t^2$, which is why δ_1 cannot be lowered much: $q(0) < 0$ requires $\delta_1 < \sqrt{3}/\pi = 0.5513$.

2. The Fourier transform of the Fejér kernel

For $\Delta > 0$, the triangle $\tau(\xi) = \Delta^{-1}(1 - |\xi|/\Delta)_+$ is the convolution of $\Delta^{-1}\mathbf{1}_{[-\Delta/2,\Delta/2]}$ with itself, whose inverse Fourier transform is $\operatorname{sinc}(\pi\Delta t)$. Hence $\int \tau(\xi) e^{-2\pi i \xi t}\, d\xi = \operatorname{sinc}^2(\pi\Delta t)$, and Fourier inversion gives the formula used in the proof of Lemma 8.

3. Values of the kernel

From (2), in 200-bit ball arithmetic,

$$K_0 = 0.91872536986556843778\ldots,$$
$$1/(\delta_1 K_0) = 1.97902646189890434914\ldots,$$
$$k(\tfrac{11}{20}) = 0.59963660635252251069\ldots,$$
$$w(\tfrac{11}{20}) = 0.35956405967797003982\ldots,$$

so that $\frac{4}{3} w(\frac{11}{20}) = 0.4794187\ldots$. The minimum of k on $[0,\infty)$ is $-0.1804\ldots$, attained near $x = 1.47$.

4. Convex trace functions

If f is convex on an interval J, then $X \mapsto \operatorname{tr} f(X)$ is convex on the Hermitian matrices with spectrum in J [4, Theorem V.2.5]. For the pinching we need it on positive semidefinite matrices, with $f = \Psi$; the averages $2^{-q}\sum_\varepsilon U_\varepsilon X U_\varepsilon$ of Lemma 4 remain positive semidefinite, so the spectrum stays in $[0,\infty)$, where Ψ is convex.

Appendix B: Verification of Proposition 11

1. The table

For $c = 0, 1, \ldots, 79999$, a lower bound $\underline{w}_c$ for w on the cell $[c/4000, (c+1)/4000]$ is computed by evaluating (2) on the cell as a ball in 128-bit arithmetic [13], taking the lower bound of $|k|$, rounding it down to a double, squaring, and rounding down again. On $[20,\infty)$ only $w \ge 0$ is used. The table occupies 640,000 bytes and has SHA-256 hash `fd9304e3...`; its first 46,000 cells form the table of hash `8ea9e020...`, used for the uniform weights. Both hashes are recorded in full in the archive.

2. Rounding

All floating-point operations of the verifier run in the rounding mode towards $-\infty$. In the first bound below every quantity is nonnegative, so every computed sum and product is a lower bound for the exact one; where the second bound needs an upper bound, it is formed as $-((-x)\,y)$ or $-((-x)-y)$ in the same mode. The target $\frac{889}{125000}$ is rounded up; every weight, stored as a quotient of integers, is rounded down, and so is $b_i/4000$, and the second bound also uses them rounded up.

3. Reduction to a compact region

If the lower ends l_i of the cells that contain the gaps satisfy $\sum_i (b_i/4000)\, l_i \ge \frac{889}{125000}$, the linear term alone settles the inequality. In particular the gap g_i is confined to the cells below C_i, the least integer with $(b_i/4000)\, C_i \ge \frac{889}{125000}$; here $C_1 = C_9 = 108{,}955$, $C_2 = C_8 = 64{,}217$, $C_3 = C_7 = 62{,}579$, $C_4 = C_6 = 50{,}638$ and $C_5 = 50{,}937$. Moreover every single gap satisfies $b_i g_i + a_{1,i-1} w(g_i) < \frac{889}{125000}$, since all other terms are nonnegative. Testing this cell by cell leaves one or two intervals of cells for each gap: $[3321, 108954]$ for g_1 and g_9; $[3521, 5742]$ and $[6057, 64205]$ for g_2 and g_8; $[3564, 5424]$ and $[6401, 61988]$ for g_3 and g_7; $[3634, 5166]$ and $[6711, 49745]$ for g_4 and g_6; and $[3650, 5120]$ and $[6768, 49946]$ for g_5. The 128 products of these intervals are the initial boxes.

4. A first lower bound on a box

A box is a product of intervals of cells $[l_i, u_i]$, $i = 1, \dots, 9$. On it, $\sum b_i g_i \geq \sum (b_i/4000)\, l_i$, and the distance $y_{i+r} - y_i$ lies in the union of the cells from $\sum l$ to $\sum (u+1) - 1$ over the r gaps involved. The minimum of $\underline{w}_c$ over that range of cells, found by a sparse table of range minima, bounds w from below. Adding these bounds with their weights gives a lower bound for F on the box.

5. A second-order bound

The first bound loses the variation of w across a box, which is of first order in its size, and near the minima of F it needs boxes of a few cells. A second table removes this loss. For $m = 0, 1, \dots, 80000$ it holds a lower bound for $w(m/4000)$ and an enclosure of $w'(m/4000)$, and for each cell a lower bound for w'' on the cell, the least of the bounds on its four quarters. By (2),

$$K'(x) = \frac{\pi}{2}\big(\operatorname{sinc}'(u_-) + \operatorname{sinc}'(u_+)\big),$$
$$K''(x) = \frac{\pi^2}{2}\big(\operatorname{sinc}''(u_-) + \operatorname{sinc}''(u_+)\big),$$

with $u_\pm = \pi x \pm 1/\sqrt{2}$, and $w' = 2KK'/K_0^2$, $w'' = 2(K'^2 + KK'')/K_0^2$. These are evaluated in 128-bit ball arithmetic, with the Taylor series of sinc' and sinc'' where $|u| \leq 1$; their terms alternate and decrease, so the first omitted term bounds the tail. As a check, the values of w' and w'' were compared with difference quotients of (2) in 240-bit arithmetic at 17,000 points, and all lie in the enclosures. The table holds 320,003 doubles and has SHA-256 hash `8915be68`$\dots$

On a box, let $c_i = \lfloor (l_i + u_i + 1)/2 \rfloor$, so that $c/4000$ is the grid point nearest its centre, and write $g_i = (c_i + \delta_i)/4000$ with $\delta_i \in [l_i - c_i,\, u_i + 1 - c_i]$. A distance $y_{i+r} - y_i$ equals $(d + \Delta)/4000$, where d is the sum of the c_j over the r gaps involved and Δ that of the δ_j. By Taylor's theorem

$$w\Big(\frac{d+\Delta}{4000}\Big) \geq w\Big(\frac{d}{4000}\Big) + w'\Big(\frac{d}{4000}\Big)\frac{\Delta}{4000} + \frac{m_2}{2}\Big(\frac{\Delta}{4000}\Big)^2,$$

where m_2 is the least lower bound for w'' over the cells that the distance can reach, found from a second sparse table of range minima. Summing with the weights, F is at least its value at the grid point, plus a linear form $\sum_i \beta_i \delta_i$, plus the quadratic terms. Here β_i collects $b_i/4000$ and the terms $a_{r,j} w'(d/4000)/4000$ of the pairs whose distance spans the gap g_i; it is known up to an interval, and $\beta_i \delta_i$ is at least the least of the four products of the ends of the two intervals. A quadratic term is at least 0 when $m_2 \geq 0$, and at least $\frac{1}{2} a_{r,j} m_2 \max(\Delta_-^2, \Delta_+^2)/4000^2$ otherwise, where $[\Delta_-, \Delta_+]$ is the range of Δ. The loss of this bound is of second order in the size of the box. It is not used for boxes that reach beyond the table.

6. The search

A box is discarded when the larger of the two bounds reaches $\frac{889}{125000}$; otherwise its widest coordinate interval is halved. The verification fails if a box consisting of single cells is not discarded; this happens for the target 0.0071125, at a box of single cells next to one of the minima shown in Fig. 5.

7. The run

The boxes are first split breadth-first, with the first bound, until 207,518 remain. These are then processed depth-first, in parallel, with the rounding mode set in each thread; a thread that runs out of boxes takes half of the stack of a busy one. The search examined 1,424,333,714 boxes in under three minutes on four threads of a four-core processor and discarded all of them. The second bound is what makes the run short: with the first bound alone and the weaker target $\frac{22}{3125}$, one sixty-fourth of the search took 88,467,547,937 boxes and almost an hour, against 33,813,175 boxes and eight seconds with both. With the first bound alone, the weights for nine points with $\mu = \frac{19999}{5000000}$ give $F \geq \frac{7}{1000}$ after 52,891,178,741 boxes, in about 30 minutes, and for eight points the weights with $\mu = \frac{12499}{5000000}$ give $F \geq \frac{19}{4000}$ after 21,067,790,614 boxes. With uniform weights the program of the first bound certifies $F_8 \geq \frac{13}{3125}$ for $\mu_0 = \frac{1}{3200}$ after 10,039,560,518 boxes, using the table of 46,000 cells. The run for $\mu_0 = \frac{1}{2000}$ and $c_0 = \frac{605}{100000}$, made with the serial version, examined 5,440,924,378 boxes. The same verifier reproduces the inequality of [1] for $R = 7$, $\mu_0 = \frac{1}{3000}$, $c_0 = \frac{19}{5000}$, with the same three intervals of cells.

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